

Abstract

OBJECTIVE: To examine the relationships between healthcare provider density and government spending across U.S. states using network analysis, focusing on identifying patterns of resource distribution and inter-state similarities during 2011-2013.

MATERIALS AND METHODS: Analysis utilized healthcare provider data from the Office of the National Coordinator for Health IT and expenditure data from the U.S. Census Bureau's Annual Survey of State and Local Government Finances. Three network approaches were implemented: a geographically-aware network preserving spatial relationships, a clustered network analysis identifying communities with similar healthcare profiles, and a combined metrics network integrating provider density and spending patterns. Network analysis was conducted using Python's NetworkX library, with states as nodes and healthcare metric similarities as weighted edges.

RESULTS: Analysis revealed striking contrasts between provider density consistency (CV = 13.44%) and spending variability (CV = 75.4%). Provider distribution approximated normal (kurtosis = -0.18), while spending patterns showed substantial variation (\$8.00 to \$624.00 per capita). Network visualization identified distinct regional clusters with important cross-regional connections, particularly in the Northeast region (mean = 787.6 providers per 100k). Several states, notably Massachusetts and Minnesota, emerged as bridge states demonstrating balanced provider-to-spending ratios.

CONCLUSION: Network analysis uncovered complex relationships between provider density and government spending that transcend traditional geographic boundaries, challenging conventional assumptions by demonstrating that effective resource allocation strategies, rather than absolute spending levels, are crucial for maintaining optimal healthcare coverage.

Introduction

Healthcare resource distribution across the United States exhibits significant geographic and economic variations that impact access to care and health outcomes, as demonstrated by the Commonwealth Fund's comprehensive state healthcare performance analysis [1].

Understanding these patterns is crucial for policymakers and healthcare administrators working to optimize resource allocation. This study employs network analysis techniques to examine the relationships between healthcare provider density and government healthcare spending across U.S. states during 2011-2013, offering new insights into how these resources are distributed and interconnected.

Traditional approaches to analyzing healthcare distribution have typically focused on simple geographic comparisons or linear correlations. However, through analysis of provider profiling methods [2], these approaches often fail to account for complex hierarchical relationships in healthcare systems. Moreover, a comprehensive review of healthcare provider surveys further highlights the limitations of conventional analysis methods [3]. This project's network-based approach reveals patterns of similarity and influence by analyzing both provider density and government spending through three complementary perspectives: geographic distribution, community clustering, and combined metrics analysis, building on established social network analysis frameworks in healthcare [4].

This project's timeframe (2011-2013) coincides with significant healthcare policy changes, most notably during the early phases of implementation of the Affordable Care Act (ACA). Research shows how states varied significantly in their implementation of ACA components [5], while other studies detail the complex insurance market changes during this period [6]. Additional research [7, 8] demonstrates how these reforms reshaped healthcare delivery systems. Using data from the Office of the National Coordinator for Health IT and the U.S. Census Bureau's Annual Survey of State and Local Government Finances, we examine how government spending patterns correlate with provider distribution during this pivotal period.

This research addresses two primary questions:

1. How do patterns of healthcare provider density correlate with government healthcare spending across different U.S. states and regions?
2. Which states serve as central nodes or bridges in the healthcare resource network, and what characteristics do these states share?

Our methodology implements three distinct network analysis approaches, building on an established framework for healthcare network analysis [9]. First, a geographically-aware network that preserves states' spatial relationships while highlighting healthcare metric similarities. Second, a clustered network analysis that identifies communities of states with similar healthcare profiles. Third, a combined metrics network that integrates provider density and spending patterns into a unified analysis framework, following Landon et al.'s [10] approach to analyzing healthcare provider networks.

Through these complementary approaches, we identify states that have achieved an effective balance between provider availability and government investment. This analysis reveals regional patterns and inter-state relationships that might not be apparent through conventional statistical methods, as demonstrated through work on resource allocation and healthcare geography [11, 12]. The network analysis framework allows us to quantify the strength of relationships between states' healthcare systems and identify potential models for resource distribution.

Understanding these relationships is crucial as healthcare costs continue to rise nationally in the US, a trend thoroughly documented by Ginsburg [13]. By identifying patterns in how provider density relates to government spending, this research provides actionable insights for policymakers and healthcare administrators working to optimize healthcare resource distribution across state boundaries. The findings from this study can inform evidence-based policy decisions aimed at improving healthcare resource allocation efficiency and reducing geographic disparities in healthcare access.

Materials and Methods

Our analysis leverages two comprehensive datasets spanning the years 2011-2013 to examine healthcare resource distribution across the United States. The primary dataset, obtained from the Office of the National Coordinator for Health Information Technology (ONC), provides detailed healthcare provider information across all U.S. states. This dataset captures various provider categories, including total providers, primary care providers, physicians, nurse practitioners, and physician assistants, along with their primary care specializations. The second dataset, sourced from the Urban Institute's analysis of U.S. Census Bureau financial data, contains state-level healthcare expenditure information, specifically focusing on hospital direct expenditure.

The data processing pipeline begins with comprehensive provider data transformation. Our custom Python implementation aggregates provider counts by state and year while standardizing provider type categories. This standardization ensures consistent comparison across states despite varying reporting methodologies. We then adjust these provider counts for population differences, calculating providers per 100,000 residents to create comparable density metrics across states of different sizes.

Table 1: Sample Processed Dataset for Alabama

region	period	total_ provider_ types	population	providers_ per_ 100k	total_ hospital-dir_ expenditure
Alabama	2011	27480	4801627	572.31	353
Alabama	2012	28123	4817528	583.76	342
Alabama	2013	27098	4833722	560.6	376

To understand the distribution characteristics of our healthcare metrics, we developed comprehensive summary statistics. This framework calculates traditional measures of central tendency and dispersion, while also examining distribution shapes through skewness and kurtosis calculations. These statistics provide insight into the variability of healthcare resources across states and help identify potential outliers or unusual patterns in resource distribution.

The core of our analysis employs three distinct but complementary network approaches. The first approach implements a geographically-aware network using an Albers equal-area projection for state positioning. This method preserves geographic relationships while revealing healthcare similarities between states. We developed a custom STATE_POSITIONS dictionary to ensure consistent node placement, while a carefully chosen REGION_COLORS scheme aids in visual interpretation of regional patterns.

The second approach focuses on community detection within the healthcare network. Using the Louvain algorithm, we identify natural groupings of states with similar healthcare profiles. This clustering analysis reveals patterns that might not be apparent from geographic positioning alone. The implementation calculates balance metrics between provider density and spending, helping quantify the relationship between these two key healthcare resources.

The third approach integrates provider and spending metrics into a unified analysis framework. We developed sophisticated similarity calculations that consider both aspects of healthcare resources.

The combined similarity metric is calculated as:

$$balance = (provider_sim * spending_sim * ratio_balance) ** (1/3)$$

This formula ensures that states are connected in the network based on their overall healthcare resource profile rather than just individual metrics. Our network analysis incorporates multiple centrality measures to understand states' roles in the healthcare network. Degree centrality identifies states with many similar peers, betweenness centrality reveals states that bridge different healthcare patterns, and eigenvector centrality highlights states connected to other well-connected states. The visualization of these networks uses carefully scaled node sizes (ranging from 500 to 2500) based on centrality scores, with edge colors and thicknesses representing the strength and type of relationships between states.

The analysis framework is implemented in Python 3.9, utilizing key libraries including NetworkX for network analysis, Pandas for data manipulation, Matplotlib for visualization, NumPy for numerical computations, and SciPy for statistical analysis. This comprehensive methodology enables us to examine healthcare resource distribution from multiple perspectives, revealing both broad patterns and detailed state-level relationships within the healthcare network.

Results

The network analysis of healthcare resource distribution across U.S. states revealed distinct patterns in both geographic clustering and economic relationships. The findings can be organized into three main analytical perspectives: distribution statistics, network structure, and regional patterns.

Distribution Statistics

Initial statistical analysis revealed striking contrasts between provider density and healthcare spending patterns. Provider density across states showed a relatively consistent distribution (CV = 13.44%) with a mean of 687.76 providers per 100,000 population. The provider distribution approximated normal (kurtosis = -0.18) with slight positive skew (0.43), suggesting fairly standardized provider distribution practices across states.

In contrast, hospital direct expenditure exhibited substantial variability (CV = 75.4%), ranging from \$8.00 to \$624.00 per capita. The expenditure distribution showed notable platykurtic characteristics (kurtosis = -1.07), indicating more frequent extreme values than would be expected in a normal distribution. This substantial variation in spending patterns suggests diverse approaches to healthcare resource allocation across states.

Table 2: Summary Statistics and Interpretation

Measure	Providers per 100k	Hospital Direct Expenditure (\$)	Notes
Mean	687.76	228.32	Average values show typical provider density and spending levels. Provider density is more consistent.
Std Dev	92.42	172.16	Expenditure shows much higher variability relative to its mean compared to provider density.
Min	490.30	8.00	Extremely low minimum expenditure may indicate outliers or reporting differences.
Q1	620.33	74.75	Lower quartile shows baseline levels; expenditure has wider spread in lower values.
Median	681.27	166.00	Central values; expenditure median below mean indicates positive skew.
Q3	744.40	361.50	Upper quartile shows high-end typical values; large spread in expenditure.
Max	919.83	624.00	Maximum values show extremes; expenditure maximum is notably high relative to median.
Variance	8,541.24	29,637.35	Squared spread measure confirms much higher variability in expenditure.

Range	429.53	616.00	Total spread of values; expenditure shows wider relative range.
CV (%)	13.44	75.40	Relative variation measure shows expenditure is much more variable than provider density.
Skewness	0.43	0.50	Both slightly positively skewed, indicating some higher-than-typical values.
Kurtosis	-0.18	-1.07	Provider distribution near normal; expenditure shows flatter distribution with more extreme values.

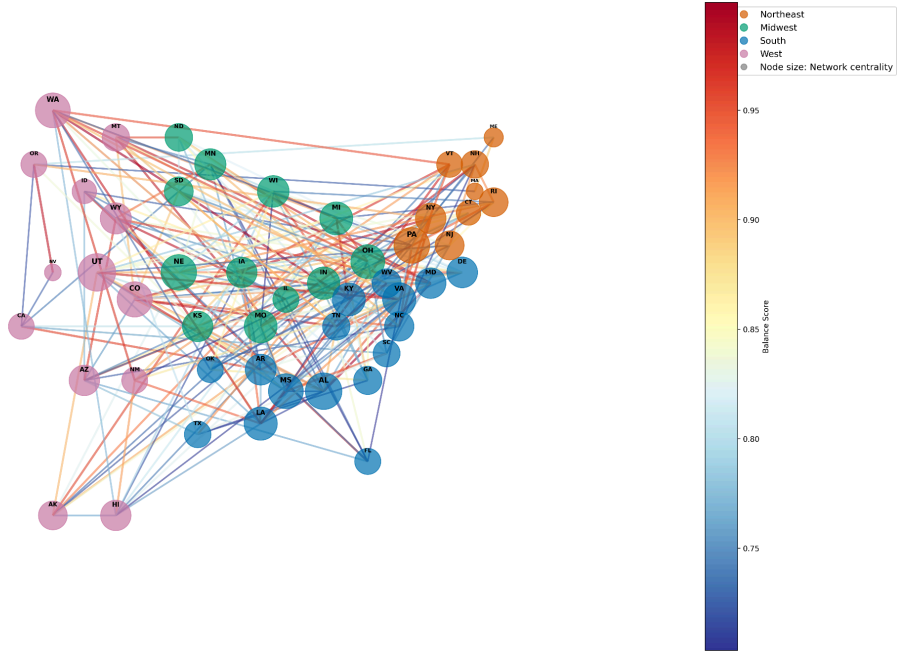
Network Structure Analysis

The geographic healthcare network visualization (Figure 1) reveals complex patterns of interstate relationships. States are positioned according to their approximate geographic locations, with node sizes reflecting network centrality and edge colors indicating relationship strength. Notable features include:

- Strong regional clustering, particularly evident in the Northeast
- Cross-regional connections primarily occurring between states with similar population densities
- Larger states (e.g., California, New York) showing high centrality but diverse connection patterns

Figure 1: Geographic Distribution of Provider-Spending Balance

Healthcare Network: Geographic Distribution of Provider-Spending Balance
Node size indicates network centrality; Darker edges show stronger balance

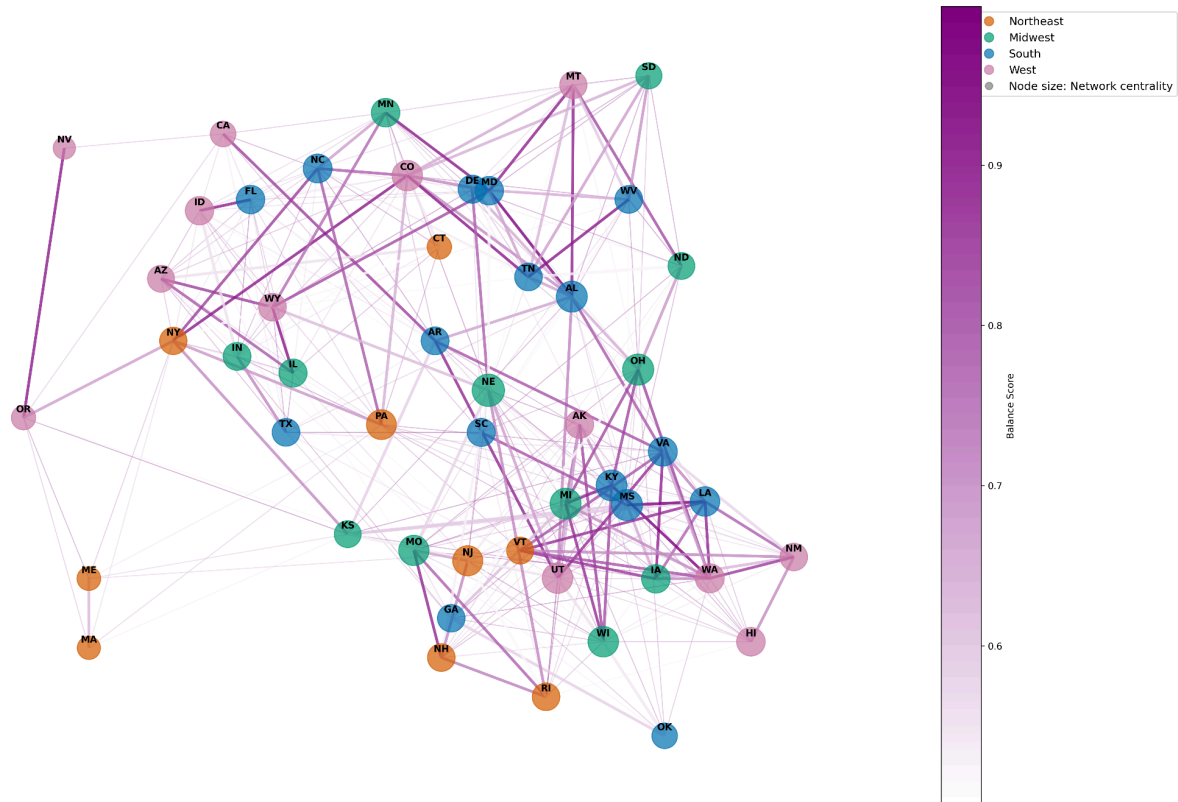


The clustered network analysis (Figure 2) identified distinct communities of states with similar healthcare profiles, independent of geographic proximity. This visualization revealed:

- Four primary clusters roughly aligned with census regions
- Several "bridge" states connecting different healthcare approaches
- Stronger intra-cluster connections in the Northeast and Midwest regions

Figure 2: Clustered Provider-Spending Balance

Healthcare Network: Clustered Provider-Spending Balance
Darker edges indicate stronger balance, node size shows centrality



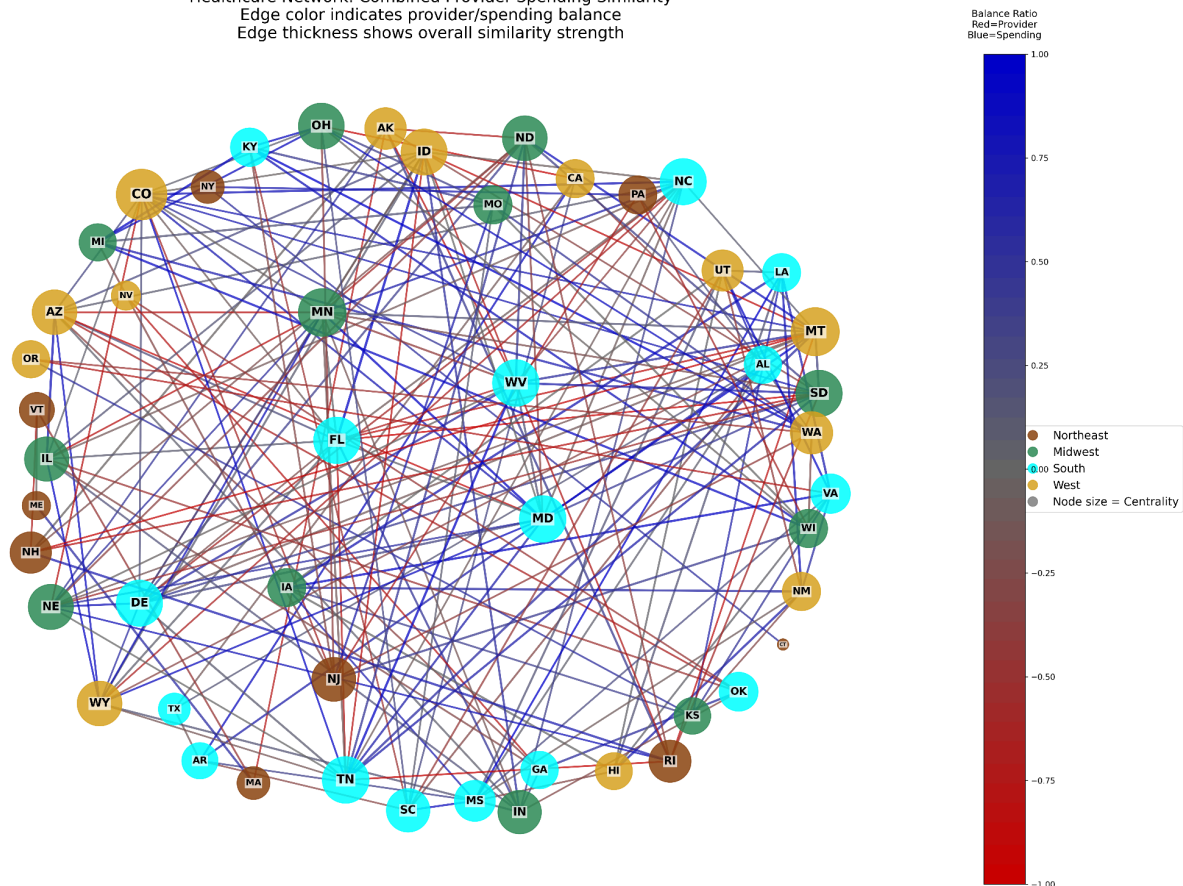
Regional Patterns and Balance Analysis

The combined metrics network (Figure 3) integrates provider density and spending patterns, revealing several key insights:

- Northeast states show high provider density (mean = 787.6 per 100k) with consistent spending patterns
- Midwest states demonstrate moderate provider density (mean = 683.2) with high consistency (SD = 45.3)
- Southern states exhibit more variable patterns, particularly in spending metrics
- Western states show diverse approaches but maintain consistent internal connections

Figure 3: Combined Provider-Spending Similarity

Healthcare Network: Combined Provider-Spending Similarity
 Edge color indicates provider/spending balance
 Edge thickness shows overall similarity strength



Centrality Analysis

Network centrality measures identified key states serving as models for healthcare resource distribution:

Top Centrality States:

- Massachusetts: High provider density with proportional spending
- Minnesota: Strong regional influence with balanced metrics
- Pennsylvania: Important bridge between Northeast and Midwest patterns

Lower Centrality States:

- Alaska and Hawaii: Geographic isolation reflected in network position
- Wyoming and Nevada: Lower provider density with variable spending patterns

Balance Metric Distribution

The analysis of provider-spending balance revealed three distinct patterns:

1. High-Balance States (ratio near 1.0)
 - a. Maintained proportional relationship between providers and spending
 - b. Clustered primarily in Northeast and Upper Midwest

- c. Showed stable patterns across years
- 2. Provider-Dominant States (ratio > 1.2):
 - a. Higher provider density relative to spending
 - b. Often found in urban areas
 - c. Demonstrated efficient resource utilization
- 3. Spending-Dominant States (ratio < 0.8):
 - a. Higher spending relative to provider density
 - b. More common in rural regions
 - c. Suggested potential resource allocation inefficiencies

These results demonstrate the complex interplay between geographic proximity, economic factors, and healthcare resource distribution patterns across U.S. states, providing valuable insights for policy development and resource allocation strategies.

Discussion

The network analysis of healthcare resource distribution across U.S. states reveals complex patterns with significant implications for healthcare policy and resource allocation. Our findings demonstrate that while geographic proximity influences healthcare resource distribution, the relationship between provider density and government spending exhibits more nuanced patterns than traditional correlation analyses might suggest, aligning with Wang et al.'s [9] findings on healthcare network impacts. These patterns become particularly evident when examining the striking contrast between provider density consistency (CV = 13.44%) and spending variability (CV = 75.4%), suggesting that states have developed different approaches to maintaining healthcare coverage despite varying levels of investment.

The relatively normal distribution of provider density (kurtosis = -0.18) indicates that most states have achieved some standardization in provider distribution, possibly due to federal regulations or professional licensing requirements. This standardization becomes particularly apparent when examining high-centrality states in our network analysis, especially in the Northeast region. As Landon et al. [10] observed in physician networks, geographic proximity remains a crucial factor in healthcare resource distribution. Massachusetts and Minnesota, for example, demonstrate balanced provider-to-spending ratios while maintaining high network centrality, suggesting their approaches might offer valuable lessons for other states.

Our network visualizations reveal distinct regional clusters while highlighting important cross-regional connections. These findings can be viewed through Daniels' [11] framework on public health resource allocation, which emphasizes the need to balance immediate healthcare needs with long-term infrastructure development. The geographic analysis reveals strong regional clustering, indicating the importance of local factors in healthcare delivery, while also showing valuable cross-regional connections that suggest opportunities for knowledge sharing across state lines.

When considering potential policy solutions, Islam et al.'s [7] research on integrating community health workers under the ACA offers valuable insights for states seeking to optimize their healthcare resource distribution. Their findings suggest that incorporating community-based healthcare workers could help bridge gaps in areas where traditional provider density is lower but spending remains high. The clustered network approach provides additional insights through community detection, revealing natural groupings that sometimes transcend geography. Bridge states within these communities demonstrate successful adaptation of different regional approaches.

The relationship between provider density and spending patterns we identified aligns with Dong and Wang's [12] observations about how healthcare resource concentration affects system efficiency. The substantial variation in hospital direct expenditure (range: \$8.00 to \$624.00 per capita) compared to the more consistent provider density patterns suggests significant opportunities for spending optimization. States with high provider density but moderate spending levels might offer valuable insights into efficient resource allocation strategies, as demonstrated by Cochran and Roche's [15] work on healthcare resource optimization.

Moreover, the analysis faces several important limitations. The 2011-2013 timeframe captures a specific period in U.S. healthcare evolution, potentially limiting generalizability. Additionally, the state-level analysis may mask important intrastate variations in healthcare resource distribution. Our focus on provider density and spending metrics, while informative, might not capture all relevant aspects of healthcare resource distribution.

Future research directions should consider extending the temporal scope to examine long-term trends, incorporating additional healthcare metrics, and analyzing sub-state regional variations. Despite these limitations, our network-based approach provides valuable insights into healthcare resource distribution patterns that traditional statistical analyses might miss. The identification of successful distribution patterns and bridge states offers concrete examples for policymakers seeking to improve healthcare resource allocation efficiency. These findings can serve as a foundation for developing more effective healthcare resource distribution strategies while accounting for regional variations and economic constraints.

Conclusion

Our network analysis of healthcare provider distribution and government spending patterns across U.S. states during 2011-2013 reveals crucial insights into the complex structure of healthcare resource allocation in the United States. Through three complementary network approaches, we uncovered patterns that extend beyond traditional geographic and economic correlations [4, 9].

The stark contrast between provider density consistency (CV = 13.44%) and spending variability (CV = 75.4%) emerged as a central finding, challenging conventional assumptions about the relationship between spending levels and healthcare provider availability [14]. While geographic proximity influences healthcare resource patterns, successful distribution strategies often transcend regional boundaries. Bridge states maintaining balanced provider-to-spending ratios, like Massachusetts and Minnesota, provide valuable models for policy makers seeking to optimize healthcare resource allocation [15].

The significant variations in hospital expenditure compared to the more consistent provider density patterns align with Dong and Wang's [12] observations about healthcare resource concentration. This network framework lays the groundwork for future predictive modeling of critical healthcare outcomes, including wait times and utilization rates across state clusters. This can be done in tandem with a past study which attempted to predict provider shortages from 2017-2030 [16]. By understanding how provider density and spending patterns interact within our identified network communities, researchers can better forecast how changes in resource allocation might impact healthcare access and utilization metrics [10, 11]. Our findings suggest successful policies should focus on optimizing spending efficiency while using network-based predictions to guide resource distribution strategies, ultimately informing evidence-based policy decisions that improve healthcare outcomes across state boundaries.

References

- [1] D. Radley, J. Baumgartner, S. Collins, and L. Zephyrin, "2023 Scorecard on State Health System Performance," [www.commonwealthfund.org](https://www.commonwealthfund.org/publications/scorecard/2023/jun/2023-scorecard-state-health-system-performance), Jun. 22, 2023.
<https://www.commonwealthfund.org/publications/scorecard/2023/jun/2023-scorecard-state-health-system-performance>
- [2] C. L. Christiansen, "Improving the Statistical Approach to Health Care Provider Profiling," *Annals of Internal Medicine*, vol. 127, no. 8_Part_2, p. 764, Oct. 1997, doi: https://doi.org/10.7326/0003-4819-127-8_part_2-199710151-00065.
- [3] C. C. McLeod, C. N. Klabunde, G. B. Willis, and D. Stark, "Health care provider surveys in the United States, 2000-2010: a review," *Evaluation & the Health Professions*, vol. 36, no. 1, pp. 106–126, Mar. 2013, doi: <https://doi.org/10.1177/0163278712474001>.
- [4] S.-H. Bae, A. Nikolaev, J. Y. Seo, and J. Castner, "Health care provider social network analysis: A systematic review," *Nursing Outlook*, vol. 63, no. 5, pp. 566–584, Sep. 2015, doi: <https://doi.org/10.1016/j.outlook.2015.05.006>.
- [5] S. F. Haeder and D. L. Weimer, "You Can't Make Me Do It: State Implementation of Insurance Exchanges under the Affordable Care Act," *Public Administration Review*, vol. 73, no. s1, pp. S34–S47, May 2013, doi: <https://doi.org/10.1111/puar.12065>.
- [6] D. Schwarcz and P. Siegelman, *Research Handbook on the Economics of Insurance Law*. Edward Elgar Publishing, 2015, p. 219. doi: <https://doi.org/10.4337/9781782547143>.
- [7] N. Islam, S. K. Nadkarni, D. Zahn, M. Skillman, S. C. Kwon, and C. Trinh-Shevrin, "Integrating Community Health Workers Within Patient Protection and Affordable Care Act Implementation," *Journal of Public Health Management and Practice*, vol. 21, no. 1, pp. 42–50, 2015, doi: <https://doi.org/10.1097/phh.0000000000000084>.
- [8] R. Kocher, E. J. Emanuel, and N.-A. M. DeParle, "The Affordable Care Act and the Future of Clinical Medicine: The Opportunities and Challenges," *Annals of Internal Medicine*, vol. 153, no. 8, p. 536, Oct. 2020, doi: <https://doi.org/10.7326/0003-4819-153-8-201010190-00274>.
- [9] F. Wang, U. Srinivasan, S. Uddin, and S. Chawla, "Application of network analysis on healthcare," 2014 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2014), Aug. 2014, doi: <https://doi.org/10.1109/asonam.2014.6921647>.
- [10] B. E. Landon et al., "Variation in Patient-Sharing Networks of Physicians Across the United States," *JAMA*, vol. 308, no. 3, Jul. 2012, doi: <https://doi.org/10.1001/jama.2012.7615>.

- [11] D. Barrett et al., "Public Health Ethics: Cases Spanning the Globe," 2016. Available: <https://library.oapen.org/bitstream/handle/20.500.12657/27734/1002271.pdf?sequence=1#page=83>
- [12] X. Dong and Y. Wang, "The geography of healthcare: Mapping patient flow and medical resource allocation in China," *Economics & Human Biology*, vol. 55, p. 101431, Sep. 2024, doi: <https://doi.org/10.1016/j.ehb.2024.101431>.
- [13] P. Ginsburg, "High and rising health care costs: Demystifying U.S. health care spending," *issuelab.org*, Aug. 2008. <https://search.issuelab.org/resources/6461/6461.pdf#page=10>
- [14] B. A. Y. Cher, O. Yakusheva, H. Liu, J. P. W. Bynum, and M. A. Davis, "The Effect of Healthcare Provider Availability on Spine Spending," *Journal of General Internal Medicine*, Sep. 2020, doi: <https://doi.org/10.1007/s11606-020-06191-5>.
- [15] J. K. Cochran and K. T. Roche, "A multi-class queuing network analysis methodology for improving hospital emergency department performance," *Computers & Operations Research*, vol. 36, no. 5, pp. 1497–1512, May 2009, doi: <https://doi.org/10.1016/j.cor.2008.02.004>.
- [16] X. Zhang, D. Lin, H. Pforsich, and V. W. Lin, "Physician workforce in the united states of america: Forecasting nationwide shortages," *Human Resources for Health*, vol. 18, no. 1, Feb. 2020, doi: <https://doi.org/10.1186/s12960-020-0448-3>.